PROJECT 04

04

# Amazon Products Market Analysis

Pricing Strategy · Best Sellers · Customer Satisfaction

Python | SQL Server | Power BI

*April 2026 · Raneem*

# Project Overview

## Business Problem

01

A company considering launching on Amazon needed data-driven insights into pricing, competition, and customer behavior.

## Dataset

02

1,242 raw Amazon product records across laptops, monitors, keyboards, mice, accessories, and notebooks.

## Outcome

03

Clean dataset of 971 products analyzed for 3 business problems, with SQL queries and Power BI dashboard.

# Data Cleaning & Preprocessing

| 1,242 | 903 | 271 | 971 |
|---|---|---|---|
| Raw products | Missing brands fixed | Rows removed | Clean products |

1 **Brand Extraction**

903 products had missing brand names. Manually identified 152 known brands and matched them against the first word of each product title. Filled 667 brands this way. The remaining 236 products with no identifiable brand were labeled as Generic.

2 **Duplicate Handling**

Identified 154 products with duplicates (up to 6 occurrences). Used duplicates to fill missing price, rating, review, and sales data before removing 215 duplicate rows.

3 **Data Type Fixes**

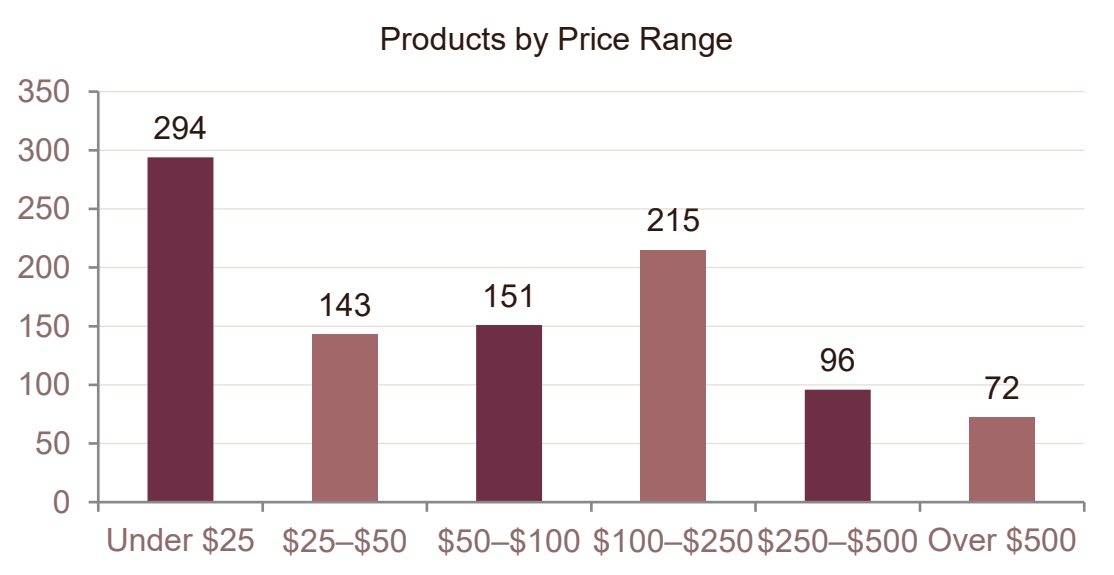
Converted price and reviews to float. Extracted numeric rating from '4.4 out of 5 stars' format. Converted sold_past_month from text (8K+) to numeric (8000). Dropped 65 rows with missing price.

# Pricing Strategy

**$139**
Average Price

**$2.49**
Minimum Price

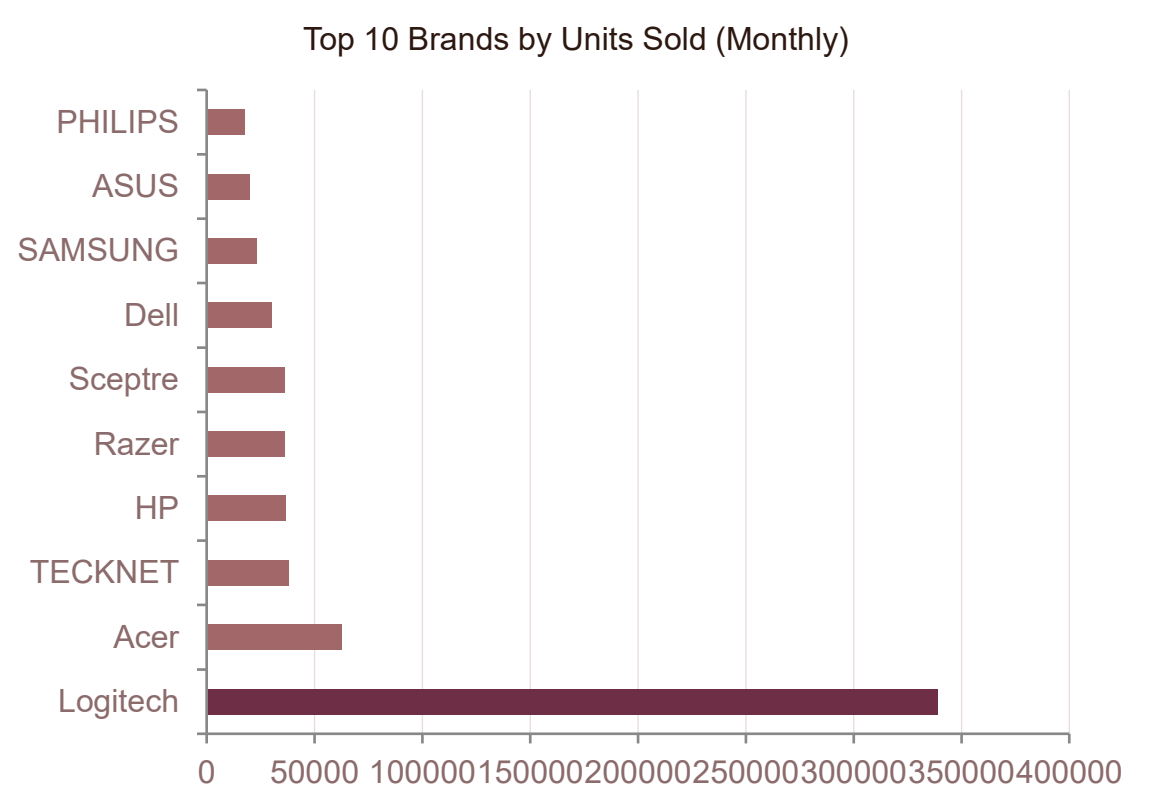
**$999**
Maximum Price




## Key Insights

- Most products (294) are priced under $25
- Mid-range $100–$250 is the 2nd largest segment
- Higher priced products have slightly lower avg ratings (4.3 vs 4.5)
- Cheaper products attract significantly more reviews

# Best Selling Products




**Logitech**

**339K** units sold

**Acer**

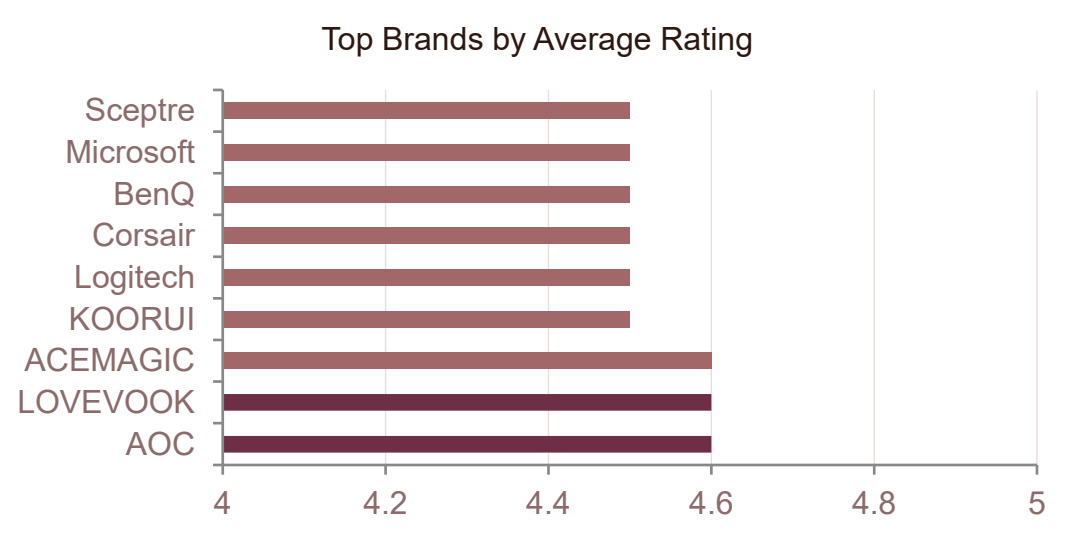
**62.6K** units sold

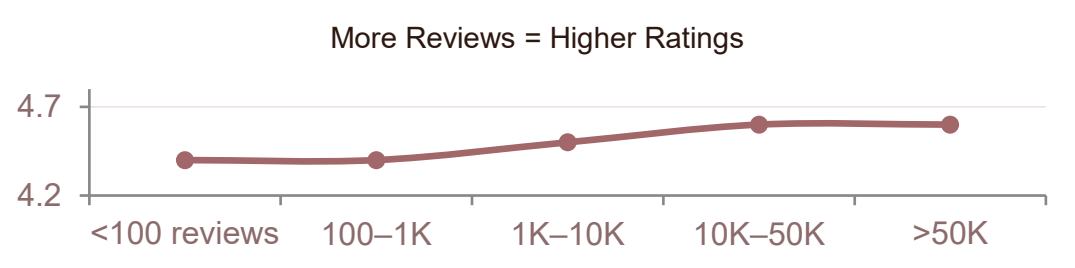
**TECKNET**

**38K** units sold

*MROCO has the highest avg sales per product at 4,767 units/month despite only 3 products.*

# Customer Satisfaction




## Key Insights

- AOC, LOVEVOOK & ACEMAGIC lead with 4.6 avg rating
- Logitech has the most reviews: 945,018 total
- Products with 50K+ reviews avg only $18 in price
- More reviews consistently correlates with higher rating

# Key Takeaways

## Pricing —

Average price is $139. Most products are under $25. Price inversely correlates with review volume.

## Sales —

Logitech dominates with 339K monthly units. Affordable products drive volume. Generic brands underperform.

## Satisfaction —

Top brands maintain 4.4–4.6 ratings. More reviews = better rating. Budget products earn more trust.

*Tools used*

Python | SQL Server | Power BI